

WEEK-9

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SESSION-17

Task1: Implement how to list active jobs, display status information, retrieve job IDs, format output, update state information, and test job listings.

```
GNU nano 8.7.1 job_listing.c *
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/wait.h>
#include <signal.h>

#define MAX_JOBS 20

typedef struct {
    int job_id;
    pid_t pid;
    char command[100];
    char state[20];
} Job;

Job jobs[MAX_JOBS];
int job_count = 0;

void update_job_states() {
    int status;

    for (int i = 0; i < job_count; i++) {
        pid_t result = waitpid(jobs[i].pid, &status, WNOHANG);

        if (result == 0) {
            strcpy(jobs[i].state, "Running");
        } else if (result == jobs[i].pid) {
            if (WIFEXITED(status)) {
                strcpy(jobs[i].state, "Completed");
            } else if (WIFSIGNALED(status)) {
                strcpy(jobs[i].state, "Terminated");
            }
        }
    }
}

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify
```

```

GNU nano 8.7.1                                job_listing.c *
printf("      SKILL WEEK 9 - TASK 1\n");
printf("      JOB LISTING SYSTEM\n");
printf("=====\\n");

while (1) {
    printf("\\nMenu:\\n");
    printf("1. Start a Job\\n");
    printf("2. List Active Jobs\\n");
    printf("3. List All Jobs\\n");
    printf("4. Update Job States\\n");
    printf("5. Exit\\n");
    printf("Enter your choice: ");

    scanf("%d", &choice);

    switch (choice) {
        case 1:
            printf("Enter job command name: ");
            scanf("%99s", command);

            printf("Enter execution time in seconds: ");
            scanf("%d", &seconds);

            create_job(command, seconds);
            break;

        case 2:
            display_jobs();
            break;

        case 3:
            display_all_jobs();
            break;

        case 4:
            update_job_states();
            printf("Job states updated successfully.\\n");

        case 5:
            break;
    }
}

^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute
^X Exit      ^R Read File  ^\\ Replace    ^U Paste      ^J Justify

```

```

saivihamsh@Varshini-PC:~$ cd ~/OSSP
saivihamsh@Varshini-PC:~/OSSP$ mkdir -p skill/week9/task1
saivihamsh@Varshini-PC:~/OSSP$ cd skill/week9/task1
pwd
/home/madhulika/OSSP/skill/week9/task1
saivihamsh@Varshini-PC:~/OSSP/skill/week9/task1$ nano job_listing.c
saivihamsh@Varshini-PC:~/OSSP/skill/week9/task1$ gcc job_listing.c -o T1
saivihamsh@Varshini-PC:~/OSSP/skill/week9/task1$ ./T1
=====
      SKILL WEEK 9 - TASK 1
      JOB LISTING SYSTEM
=====

Menu:
1. Start a Job
2. List Active Jobs
3. List All Jobs
4. Update Job States
5. Exit
Enter your choice: 1
Enter job command name: slepp
Enter execution time in seconds: 10

Job started successfully.
Job ID : 1
PID    : 674
Command: slepp

Menu:
1. Start a Job
2. List Active Jobs
3. List All Jobs
4. Update Job States
5. Exit
Enter your choice: 1
Enter job command name: backup
Enter execution time in seconds: 15

Job started successfully.
Job ID : 2
PID    : 685

```

```

Command: backup

Menu:
1. Start a Job
2. List Active Jobs
3. List All Jobs
4. Update Job States
5. Exit
Enter your choice: 2

===== ACTIVE JOBS =====
Job ID   PID      Status    Command
-----
2        685      Running   backup
=====

Menu:
1. Start a Job
2. List Active Jobs
3. List All Jobs
4. Update Job States
5. Exit
Enter your choice: 5
Exiting Job Listing System.

```

Observation: The job listing system was successfully implemented using C and process management functions. The program creates and maintains a list of jobs with unique job IDs and process IDs, displays active jobs along with their status and command information, and updates the state of jobs as they complete. The formatted output makes the job information easy to understand, and the program was tested by creating multiple jobs and listing their current states successfully.

Task2: To design how to Identify Target Job, Transfer Terminal Control, Resume Processes, Wait for Completion, Update Job State, Test Foreground Switching.

```
GNU nano 8.7.1 job_control.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <signal.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <termios.h>

#define MAX_JOBS 10

typedef enum {
    RUNNING,
    STOPPED,
    COMPLETED
} JobState;

typedef struct {
    int job_id;
    pid_t pid;
    JobState state;
} Job;

Job jobs[MAX_JOBS];
int job_count = 0;

pid_t shell_pgid;
int terminal_fd;

void display_state(JobState state)
{
    if (state == RUNNING)
        printf("Running");
    else if (state == STOPPED)
        printf("Stopped");
    else
        printf("Completed");
}

^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify
```

```
GNU nano 8.7.1 job_control.c *
void create_job()
{
    if (job_count >= MAX_JOBS) {
        printf("Maximum number of jobs reached.\n");
        return;
    }

    pid_t pid = fork();

    if (pid < 0) {
        perror("fork");
        return;
    }

    if (pid == 0) {
        /*
         * Create a new process group for the child.
         */
        setpgid(0, 0);

        printf("\nChild process started.\n");
        printf("PID: %d\n", getpid());

        /*
         * Stop the process so that the parent can
         * demonstrate resuming and foreground control.
         */
        printf("Child process is stopping...\n");
        raise(SIGSTOP);

        /*
         * Execution resumes here after SIGCONT.
         */
        printf("Child process resumed in foreground.\n");

        sleep(3);
    }
}

^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify
```

```
GNU nano 8.7.1 job_control.c *
printf("1. Create Job\n");
printf("2. List Jobs\n");
printf("3. Bring Job to Foreground\n");
printf("4. Exit\n");
printf("=====\n");

printf("Enter your choice: ");

int choice;
scanf("%d", &choice);

switch (choice) {

    case 1:
        create_job();
        break;

    case 2:
        display_jobs();
        break;

    case 3:
        foreground_job();
        break;

    case 4:
        printf("Exiting Job Control System.\n");
        return 0;

    default:
        printf("Invalid choice.\n");

}

return 0;
}
```

^G Help **^O** Write Out **^F** Where Is **^K** Cut **^T** Execute
^X Exit **^R** Read File **^N** Replace **^U** Paste **^J** Justify

```
saivihamsh@Varshini-PC:~/OSSP/skill/week9$ mkdir -p task2
saivihamsh@Varshini-PC:~/OSSP/skill/week9$ cd task2
saivihamsh@Varshini-PC:~/OSSP/skill/week9/task2$ pwd
/home/madhulika/OSSP/skill/week9/task2
saivihamsh@Varshini-PC:~/OSSP/skill/week9/task2$ nano job_control.c
saivihamsh@Varshini-PC:~/OSSP/skill/week9/task2$ gcc job_control.c -o T2
saivihamsh@Varshini-PC:~/OSSP/skill/week9/task2$ ./T2
=====
SKILL WEEK 9 - TASK 2
FOREGROUND JOB CONTROL
=====
Shell PID : 966
Shell PGID: 966

===== MENU =====
1. Create Job
2. List Jobs
3. Bring Job to Foreground
4. Exit
=====
Enter your choice: 1

Job created successfully.
Job ID : 1
PID : 971
State : Stopped

===== MENU =====
1. Create Job
2. List Jobs
3. Bring Job to Foreground
4. Exit
=====
Enter your choice:
Child process started.
PID: 971
Child process is stopping...
2
```

```

===== JOB LIST =====
Job ID   PID      State
-----
1        971      Stopped
=====

===== MENU =====
1. Create Job
2. List Jobs
3. Bring Job to Foreground
4. Exit
=====
Enter your choice: 3

===== JOB LIST =====
Job ID   PID      State
-----
1        971      Stopped
=====

Enter Job ID to bring to foreground: 1

Target Job Identified
Job ID : 1
PID    : 971
Transferring terminal control...
Resuming target process...
Waiting for foreground job...
Child process resumed in foreground.
Child process completed.
Restoring terminal control to shell...

```

Observation: The foreground job control system was successfully implemented using C and Linux process management functions. The program identified the target job using its job ID and process ID, transferred terminal control to the selected process, resumed the stopped process using SIGCONT, and waited for its completion using waitpid(). After the process completed, the terminal control was returned to the shell and the job state was updated to Completed. The foreground switching functionality was tested successfully, demonstrating proper job and terminal control.

SESSION-18

Task1: To Analyze Signal Delivery, Understand Signal Masks, Review Process Groups, Explore Terminal Signals, Test Signal Behavior.

```
GNU nano 8.7.1 signal_behavior.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <signal.h>
#include <sys/wait.h>
#include <sys/types.h>

volatile sig_atomic_t signal_received = 0;

/* Signal handler */
void signal_handler(int sig)
{
    signal_received = sig;

    if (sig == SIGUSR1)
        printf("\nChild: SIGUSR1 received successfully.\n");
    else if (sig == SIGTERM)
        printf("\nChild: SIGTERM received. Terminating...\n");
}

/* Display signal mask */
void display_signal_mask()
{
    sigset_t mask;

    sigprocmask(SIG_BLOCK, NULL, &mask);

    printf("\nCurrent Signal Mask:\n");

    if (sigismember(&mask, SIGINT))
        printf("SIGINT : BLOCKED\n");
    else
        printf("SIGINT : NOT BLOCKED\n");

    if (sigismember(&mask, SIGTERM))
        printf("SIGTERM : BLOCKED\n");
    else
        printf("SIGTERM : NOT BLOCKED\n");
}

^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify
```

```
GNU nano 8.7.1 signal_behavior.c *
    perror("fork");
    return 1;
}

if (pid == 0) {
    child_process();
}

printf("\nParent Process\n");
printf("-----\n");
printf("Child PID : %d\n", pid);

sleep(2);

printf("\nSending SIGUSR1 to child...\n");

if (kill(pid, SIGUSR1) == -1)
    perror("kill");

sleep(2);

printf("\nSending SIGTERM to child...\n");

if (kill(pid, SIGTERM) == -1)
    perror("kill");

waitpid(pid, NULL, 0);

printf("\nChild process has terminated.\n");

printf("\n=====");
printf("Signal delivery and signal behavior tested.\n");
printf("=====");

return 0;
}

^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify
```

```

saiviamsh@Varshini-PC:~/OSSP/skill/week9/task1$ nano signal_behavior.c
saiviamsh@Varshini-PC:~/OSSP/skill/week9/task1$ gcc signal_behavior.c -o
T1_signal
saiviamsh@Varshini-PC:~/OSSP/skill/week9/task1$ ./T1_signal
=====
          SKILL WEEK 9 - TASK 1
          SIGNAL BEHAVIOR
=====

Parent PID : 1157
Parent PGID: 1157

Parent Process
-----
Child PID : 1158

Child Process Information
-----
Child PID   : 1158
Parent PID  : 1157
Process Group ID : 1157

SIGINT has been blocked in the child.

Current Signal Mask:
SIGINT   : BLOCKED
SIGTERM  : NOT BLOCKED
SIGUSR1  : NOT BLOCKED

Child waiting for signals...

Sending SIGUSR1 to child...

Child: SIGUSR1 received successfully.

Sending SIGTERM to child...

Child: SIGTERM received. Terminating...

Child process finished.

```

Observation: The program successfully demonstrated Linux signal handling and process behavior using parent and child processes. Signal delivery was tested by sending SIGUSR1 and SIGTERM from the parent process to the child process. The signal mask was examined, and SIGINT was blocked in the child process to demonstrate signal masking. The process IDs and process group IDs were also displayed, showing the relationship between processes and their groups. The child process received the required signals and terminated successfully, demonstrating signal delivery, signal masks, process groups, and terminal signal behavior.

Task2: To implement how to resume Stopped Jobs, Send Continue Signals, Update Job Status, Maintain Job Table, Verify Execution, Test Background Recovery.

```
saiviamsh@Varshini-PC:~/OSSP/skill/week9/task1$ cd ~/OSSP/skill/week9
saiviamsh@Varshini-PC:~/OSSP/skill/week9$ mkdir -p task2
saiviamsh@Varshini-PC:~/OSSP/skill/week9$ cd task2
saiviamsh@Varshini-PC:~/OSSP/skill/week9/task2$ nano background_recovery.c
saiviamsh@Varshini-PC:~/OSSP/skill/week9/task2$ gcc background_recovery.c -o T2_background
saiviamsh@Varshini-PC:~/OSSP/skill/week9/task2$ ./T2_background
```

```
=====
                SKILL WEEK 9 - TASK 2
                BACKGROUND JOB RECOVERY
=====
```

```
===== MENU =====
1. Create Stopped Job
2. List Job Table
3. Resume Stopped Job
4. Verify Execution
5. Exit
```

```
=====
Enter your choice: 1
```

```
Stopped job created successfully.
Job ID : 1
PID    : 1275
Status : Stopped
```

```
===== MENU =====
1. Create Stopped Job
2. List Job Table
```

```
3. Resume Stopped Job
4. Verify Execution
5. Exit
```

```
=====
Enter your choice:
```

```
Child process started.
PID: 1275
Child process is stopping...
```

```
4
===== JOB TABLE =====
Job ID      PID      Status
-----
1           1275     Running
=====
```

```
===== MENU =====
1. Create Stopped Job
2. List Job Table
3. Resume Stopped Job
4. Verify Execution
5. Exit
```

```
=====
Enter your choice: 5
```

```
Exiting Background Job Recovery System.
saiviamsh@Varshini-PC:~/OSSP/skill/week9/task2$
```

```
GNU nano 8.7.1 background_recovery.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <signal.h>
#include <sys/wait.h>
#include <string.h>

#define MAX_JOBS 10

typedef enum {
    STOPPED,
    RUNNING,
    COMPLETED
} JobState;

typedef struct {
    int job_id;
    pid_t pid;
    JobState state;
} Job;

Job jobs[MAX_JOBS];
int job_count = 0;

void display_state(JobState state)
{
    if (state == STOPPED)
        printf("Stopped");
    else if (state == RUNNING)
        printf("Running");
    else
        printf("Completed");
}

void update_job_status(int index)
{
    int status;

    ^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute
    ^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify
```

```
GNU nano 8.7.1 background_recovery.c *
    printf("4. Verify Execution\n");
    printf("5. Exit\n");
    printf("=====\n");

    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice) {
        case 1:
            create_stopped_job();
            break;

        case 2:
            display_jobs();
            break;

        case 3:
            resume_job();
            break;

        case 4:
            display_jobs();
            break;

        case 5:
            printf("Exiting Background Job Recovery System.\n");
            return 0;

        default:
            printf("Invalid choice.\n");
    }
}

return 0;
}
```

```
^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify
```

Observation: The background job recovery system was successfully implemented and tested using Linux process control functions. A stopped job was created and maintained in the job table with its job ID, process ID, and current status. The stopped job was resumed by sending the SIGCONT signal, after which its status was updated to Running. The execution was verified through the job table, and after completion the job status changed to Completed. Thus, stopped jobs were successfully resumed and recovered in the background.